Name

$M\#$

Exam Tuesday

Models

Linear Algebra Review

We are going to learn how to write down the general solution of

$$y' = A\,y$$

from the eigenvalues and eigenvectors of the matrix A.

Just a reminder

- $A\,v_i = \lambda_i\,v_i$ with $\|v_i\| \neq 0$ defines an eigenvalue λ_i and associated eigenvector v_i.
- Eigenvalues and eigenvectors can be complex.
- Almost all $n \times n$ matrices have an eigenvector basis $\{v_1, v_2, \ldots, v_n\}$. $V = [v_1 \mid v_2 \mid \ldots \mid v_n]$ gives $V^{-1}.A.V = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$. This is diagonalization.

In[19]:=
```
n = 4;
A = RandomReal[{-1, 1}, {n, n}];
{λs, V} = Eigensystem[A]; V = Vᵀ;
λs
MatrixForm[Inverse[V].A.V]
```

Out[22]=
```
{-0.900268, 0.774685, -0.12763, 0.0371144}
```

Out[23]//MatrixForm=

$$\begin{pmatrix} -0.900268 & -8.88178\times10^{-16} & 5.55112\times10^{-16} & -5.55112\times10^{-16} \\ -2.22045\times10^{-16} & 0.774685 & -1.38778\times10^{-16} & 1.66533\times10^{-16} \\ -6.73073\times10^{-16} & 2.7929\times10^{-16} & -0.12763 & 3.81639\times10^{-17} \\ -7.42462\times10^{-16} & 2.08167\times10^{-16} & 8.32667\times10^{-17} & 0.0371144 \end{pmatrix}$$

Linear Algebra Review and ODE $y' = A\,y$

Suppose I have $A\,v_1 = \lambda_1\,v_1$ then $y_1(t) = c_1\,e^{\lambda_1 t}\,v_1$ solves $y' = A\,y$.

- All the eigen-pairs $A\,v_i = \lambda_i\,v_i$ give solutions $y_i(t) = c_i\,e^{\lambda_i t}\,v_i$.
- $y' = A\,y$ is linear so $y(t) = \sum_{i=1}^{n} c_i\,e^{\lambda_i t}\,v_i$ is an n parameter family of solutions.

- If the eigen-vectors are Linearly Independent (LI) then there are unique coefficients $c = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$ satisfying $y(0) = \sum_{i=1}^{n} c_i e^{\lambda_i 0} v_i = y_0$. All we do is solve $V c = y_0$.
- Summary. The solution of $y' = A y$ with $y(0) = y_0$ is $y(t) = \sum_{i=1}^{n} c_i e^{\lambda_i t} v_i$ with $V c = [v_1 \mid \ldots \mid v_n] c = y_0$